IPCON 耗材指南

选购耗材

本耗材指南对 IPCON 耗材的性能、打印要求和应用进行了全面比较，旨在帮助用户选择最匹配需求的材料

如需了解详细技术参数，请在产品页面上下载 技术数据表 (TDS)

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www.ipconpolymer.com ↗

条长 = 行内相对水平（按数值）　主题色：● 强度 ● 刚度 ● 韧性/层粘 ● 热学 ● 耐湿 ● 硬度 ● 轻量 ● 流变　●●●○○ 打印难度　必须 推荐 无需　温度 / 速度 / 风扇范围　底色 = 本行主题色深浅（越深越优）　细条 = 相对表现（越长越好）

	耗材	TPU 95A HS 乳白	TPU GF 70D 沧浪	TPU CF 70D 黑色	PLA Matte 黝紫	PLA FC 哑蓝	PLA CF 元青	PETG Metal 光刃银	PETG GF 玄铁黑	PETG CF 玄铁银	PET DET & FC 蓝色
特性	**强度** 弯曲强度 - XY 方向	27 MPa（拉伸）	14 MPa（不断）	22 MPa（不断）	67 MPa	64 MPa	89 MPa	65 MPa	78 MPa	84 MPa	63 MPa
	强度 弯曲强度 - Z 方向	22 MPa（拉伸）	12 MPa（不断）	14 MPa（不断）	42 MPa	40 MPa	68 MPa	59 MPa	53 MPa	57 MPa	41 MPa
	刚度 弯曲模量 - XY 方向	—	280 MPa	427 MPa	3330 MPa	2933 MPa	4429 MPa	1955 MPa	2934 MPa	3488 MPa	1752 MPa
	刚度 弯曲模量 - Z 方向	—	235 MPa	261 MPa	2759 MPa	2693 MPa	2663 MPa	1841 MPa	2132 MPa	2376 MPa	1667 MPa
	韧性 冲击强度 - XY 方向	> 125.0 kJ/m²（不断） ▲	> 124.0 kJ/m²（不断）	> 124.0 kJ/m²（不断）	23.8 kJ/m²	29.6 kJ/m²	21.9 kJ/m²	76.8 kJ/m²	21.1 kJ/m²	37.1 kJ/m²	47.8 kJ/m²
	层间粘接 冲击强度 - Z 方向	> 125.0 kJ/m² ▲	18.2 kJ/m²	41.4 kJ/m²	5.2 kJ/m²	6.1 kJ/m²	9.0 kJ/m²	20.8 kJ/m²	10.4 kJ/m²	12.9 kJ/m²	7.3 kJ/m²
	耐热性 热变形温度，0.45 MPa	—	—	—	56 °C	55 °C	55.4 °C	68 °C	74 °C	74 °C	71 °C
	长期耐热性 最高连续使用温度 （与负载、尺寸、形状、壁厚、含水率等有关）	70 - 80 °C	70 - 90 °C	70 - 90 °C	45 - 50 °C （退火后的通常高于未退火的）	45 - 50 °C （退火后的通常高于未退火的）	45 - 53 °C （退火后的通常高于未退火的）	60 - 65 °C	60 - 68 °C	60 - 68 °C	60 - 65 °C
	饱和吸水率 25 °C，55% RH	1.04%	1.13%	1.16%	0.43%	0.40%	0.42%	0.34%	0.31%	0.32%	0.37%
	收缩率 (用特定模型与工艺测得，详见 TDS)	0.18% - 0.37%	0.26% - 0.45%	0.13% - 0.42%	0.23% - 0.41%	0.03% - 0.30% ▲	0.13% - 0.38%	0.26% - 0.51%	0.17% - 0.37%	0.08% - 0.41%	0.28% - 0.51%
	表面硬度 邵氏硬度	95A	~ 70D	~ 70D	80D	81D	81D	75D	77D	78D	78D
	密度	1.21 g/cm³	1.19 g/cm³	1.09 g/cm³	1.39 g/cm³	1.39 g/cm³	1.28 g/cm³	1.26 g/cm³	1.32 g/cm³	1.28 g/cm³	1.39 g/cm³
	熔融指数 (测试条件见 TDS)	28.4 g/10 min	10.2 g/10 min	3.5 g/10 min	17.4 g/10 min	12.0 g/10 min	8.0 g/10 min	25.8 g/10 min	19.0 g/10 min	17.5 g/10 min	25.9 g/10 min
前处理	使用前是否干燥	全系列通用 拆包后配干燥剂 + 密封盒直接使用 受潮后需要二次干燥									
	干燥条件	70–90 °C 8–12 h 鼓风干燥箱	80–100 °C 8–12 h 鼓风干燥箱	80–100 °C 8–12 h 鼓风干燥箱	55–60 °C 6–8 h 鼓风干燥箱	55–60 °C 6–8 h 鼓风干燥箱	55–70 °C 8–12 h 鼓风干燥箱	60–65 °C 6–8 h 鼓风干燥箱	65–70 °C 6–8 h 鼓风干燥箱	65–70 °C 6–8 h 鼓风干燥箱	65–70 °C 6–8 h 鼓风干燥箱
	是否兼容 AMS 或 CFS	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓
	喷嘴尺寸/材料	0.2 0.4 0.6 0.8 全材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 全材质	0.2 0.4 0.6 0.8 不锈钢、硬化钢、碳化钨、金刚石等含重金属风险很低的材质（详询喷嘴厂家）	0.2 0.4 0.6 0.8 硬化钢等加硬喷嘴	0.2 0.4 0.6 0.8 全材质	0.2 0.4 0.6 0.8 硬化钢等加硬喷嘴	0.2 0.4 0.6 0.8 硬化钢等加硬喷嘴	0.2 0.4 0.6 0.8 不锈钢、硬化钢、碳化钨、金刚石等含重金属风险很低的材质（详询喷嘴厂家）
	适用的打印底板和床温	光面 PEI / 纹理 PEI 30 – 40 °C	光面 PEI / 纹理 PEI 35 – 50 °C	光面 PEI / 纹理 PEI 35 – 50 °C	光面 PEI / 纹理 PEI 35 – 65 °C	光面 PEI / 纹理 PEI 35 – 65 °C	全类型 包括低温 / 高温 / 纹理PEI 30 – 50 °C	光面 PEI / 纹理 PEI 65 – 75 °C	光面 PEI / 纹理 PEI 65 – 75 °C	光面 PEI / 纹理 PEI 65 – 75 °C	光面 PEI / 纹理 PEI 65 – 80 °C
	粘接方式	若粘结不牢，可喷专用液体胶水 / 涂艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	若粘结不牢，可喷专用液体胶水 / 涂艾普创固体胶棒	不涂胶 / 涂食品级胶（食品接触用途） / 专用喷胶、PVP固体胶（非食品接触用途）	若粘结不牢，可喷专用液体胶水 / 涂艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	不涂胶 / 涂食品级胶（食品接触用途） / 专用喷胶、PVP固体胶（非食品接触用途）
打印设置	打印难度	●●●○○ 中等	●●●○○ 中等	●●●○○ 中等	●○○○○ 容易	●○○○○ 容易	●●○○○ 较易	●●○○○ 较易	●●○○○ 较易	●●○○○ 较易	●●●○○ 中等
	是否需要封箱打印	无需	无需	无需	无需	无需	无需	无需	无需	无需	无需
	是否使用干燥剂密封防潮	必须	必须	必须	推荐 (可获得更好打印质量)	推荐 (可获得更好打印质量)	推荐 (可获得更好打印质量)	必须	必须	必须	必须
	打印速度	最高达 300 mm/s	最高达 250 mm/s	最高达 250 mm/s	最高达 300 mm/s	最高达 300 mm/s	最高达 300 mm/s	最高达 300 mm/s	最高达 300 mm/s	最高达 250 mm/s	最高达 300 mm/s
	喷嘴温度	220 - 240 °C	210 - 250 °C	210 - 250 °C	210 - 240 °C	190 - 230 °C	210 - 240 °C	240 - 270 °C	245 - 270 °C	245 - 270 °C	270 - 300 °C
	部件冷却风扇	80% - 100%	10% - 90%	10% - 90%	60% - 100%	60% - 100%	60% - 100%	10% - 90%	10% - 90%	10% - 90%	10% - 80%
后处理	退火	无需退火 必要时 ≤ 80 °C · 6 h	无需退火 必要时 ≤ 100 °C · 6 h	无需退火 必要时 ≤ 100 °C · 6 h	45 °C · 2–4 h 50 °C · 2–4 h 55 °C · 4–6 h 上限 ≤ 60 °C	45 °C · 2–4 h 50 °C · 2–4 h 55 °C · 4–8 h 上限 ≤ 60 °C	45 °C · 2–4 h 50 °C · 2–4 h 55 °C · 4–6 h 上限 ≤ 60 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h 上限 ≤ 70 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h 上限 ≤ 70 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h 上限 ≤ 70 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h 上限 ≤ 75 °C

	耗材	PP GF 灰色	PP CF 黑色	PET GF 蓝色	PET CF 黑色	ASA GF 磨砂蓝	ASA CF 黑色	PC FR HS 苍苍色	PC GF 石榴裙	PC CF 黑色	CoPA HS 红色
特性	**强度** 弯曲强度 - XY 方向	53 MPa	46 MPa	113 MPa	134 MPa	76 MPa	80 MPa	101 MPa	108 MPa	116 MPa	113 MPa
	强度 弯曲强度 - Z 方向	31 MPa	30 MPa	75 MPa	62 MPa	49 MPa	45 MPa	75 MPa	62 MPa	73 MPa	65 MPa
	刚度 弯曲模量 - XY 方向	1464 MPa	1237 MPa	4067 MPa	5162 MPa	2758 MPa	3171 MPa	2521 MPa	3367 MPa	5482 MPa	3068 MPa
	刚度 弯曲模量 - Z 方向	664 MPa	629 MPa	2511 MPa	3136 MPa	2156 MPa	2173 MPa	2754 MPa	1948 MPa	2362 MPa	2385 MPa
	韧性 冲击强度 - XY 方向	46.5 kJ/m²	56.8 kJ/m²	40.7 kJ/m²	22.5 kJ/m²	23.5 kJ/m²	25.7 kJ/m²	62.6 kJ/m²	34.2 kJ/m²	23.7 kJ/m²	64.3 kJ/m²
	层间粘接 冲击强度 - Z 方向	13.4 kJ/m²	23.3 kJ/m²	15.4 kJ/m²	9.1 kJ/m²	8.2 kJ/m²	7.7 kJ/m²	11.2 kJ/m²	12.4 kJ/m²	16.1 kJ/m²	13.3 kJ/m²
	耐热性 热变形温度，0.45 MPa	0.45 MPa 未退火：98 °C 0.45 MPa 退火后：104 °C 1.80 MPa 未退火：84 °C 1.80 MPa 退火后：89 °C	0.45 MPa 未退火：113 °C 0.45 MPa 退火后：115 °C 1.80 MPa 未退火：81 °C 1.80 MPa 退火后：87 °C	0.45 MPa 未退火：74 °C 0.45 MPa 退火后：99 °C 1.80 MPa 未退火：66 °C 1.80 MPa 退火后：73 °C	0.45 MPa 未退火：138 °C 0.45 MPa 退火后：155 °C 1.80 MPa 未退火：76 °C 1.80 MPa 退火后：87 °C	98 °C	99 °C	92 °C	109 °C	113 °C	0.45 MPa 未退火：150 °C 0.45 MPa 退火后：152 °C 1.80 MPa 未退火：65 °C 1.80 MPa 退火后：70 °C
	长期耐热性 最高连续使用温度 （与负载、尺寸、形状、壁厚、含水率等有关）	75 - 85 °C （退火后的通常高于未退火的）	75 - 85 °C （退火后的通常高于未退火的）	65 - 75 °C （退火后的通常高于未退火的）	65 - 80 °C （退火后的通常高于未退火的）	75 - 85 °C	75 - 85 °C	70 - 85 °C	85 - 100 °C	85 - 100 °C	60 - 80 °C（干态） （退火后的通常高于未退火的）
	饱和吸水率 25 °C，55% RH	< 0.06%	< 0.06%	0.36%	0.38%	0.31%	0.33%	0.28%	0.22%	0.21%	2.98%
	收缩率 (用特定模型与工艺测得，详见 TDS)	0.25% - 0.57%	0.19% - 0.59%	0.23% - 0.43%	0.17% - 0.46%	0.13% - 0.37%	0.14% - 0.32%	0.39% - 0.50%	0.29% - 0.47%	0.21% - 0.44%	0.28% - 0.49%
	表面硬度 邵氏硬度	70D	70D	85D	86D	80D	81D	84D	85D	85D	84D
	密度	1.04 g/cm³	0.92 g/cm³ ▲	1.36 g/cm³	1.26 g/cm³	1.11 g/cm³	1.04 g/cm³	1.22 g/cm³	1.30 g/cm³	1.24 g/cm³	1.14 g/cm³
	熔融指数 (测试条件见 TDS)	11.9 g/10 min	10.8 g/10 min	15.4 g/10 min	17.8 g/10 min	19.5 g/10 min	17.7 g/10 min	19.4 g/10 min	16.4 g/10 min	14.7 g/10 min	16.3 g/10 min
前处理	使用前是否干燥										
	干燥条件	70–100 °C 6–8 h 鼓风干燥箱	70–100 °C 6–8 h 鼓风干燥箱	80–110 °C 6–10 h 鼓风干燥箱	80–110 °C 6–10 h 鼓风干燥箱	70–90 °C 4–8 h 鼓风干燥箱	70–90 °C 4–8 h 鼓风干燥箱	65–75 °C 8–12 h 鼓风干燥箱	80–100 °C 8–12 h 鼓风干燥箱	80–100 °C 8–12 h 鼓风干燥箱	100–120 °C 8–14 h 鼓风干燥箱
	是否兼容 AMS 或 CFS	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	喷嘴尺寸/材料	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 全材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 全材质
	适用的打印底板和床温	光面 PEI / 纹理 PEI 55 – 75 °C	光面 PEI / 纹理 PEI 55 – 75 °C	光面 PEI / 纹理 PEI 70 – 90 °C	光面 PEI / 纹理 PEI 70 – 90 °C	光面 PEI / 纹理 PEI 90 – 110 °C	光面 PEI / 纹理 PEI 90 – 110 °C	光面 PEI / 纹理 PEI 85 – 100 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 90 – 110 °C
	粘接方式	PP 专用胶水 / 贴膜 / 底板	PP 专用胶水 / 贴膜 / 底板	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	艾普创固体胶棒	艾普创固体胶棒	艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒
打印设置	打印难度	●●●●○ 较难*	●●●●○ 较难*	●●●○○ 中等*	●●●○○ 中等*	●●●○○ 中等	●●●○○ 中等	●●●●○ 较难	●●●●○ 较难	●●●●○ 较难	●●●●○ 较难*
	是否需要封箱打印	无需	无需	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）
	是否使用干燥剂密封防潮	必须	必须	必须	必须	必须	必须	必须	必须	必须	必须
	打印速度	最高达 250 mm/s	最高达 220 mm/s	最高达 250 mm/s	最高达 300 mm/s	最高达 300 mm/s	最高达 300 mm/s	最高达 350 mm/s	最高达 350 mm/s	最高达 350 mm/s	最高达 300 mm/s
	喷嘴温度	220 - 260 °C	220 - 260 °C	265 - 295 °C	265 - 295 °C	265 - 290 °C	265 - 290 °C	240 - 270 °C	280 - 300 °C	280 - 300 °C	270 - 300 °C
	部件冷却风扇	10% - 70%	10% - 70%	10% - 40%	10% - 40%	10% - 40%	10% - 40%	0% - 60%	0% - 40%	0% - 40%	0% - 40%
后处理	退火	60 °C · 3–4 h 70 °C · 3–4 h 80 °C · 6–8 h 上限 ≤ 90 °C	60 °C · 3–4 h 70 °C · 3–4 h 80 °C · 6–8 h 上限 ≤ 90 °C	60 °C · 3–4 h 80 °C · 3–4 h 90 °C · 6–8 h 上限 ≤ 100 °C	60 °C · 3–4 h 80 °C · 3–4 h 90 °C · 6–8 h 上限 ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h 上限 ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h 上限 ≤ 100 °C	55 °C · 2–4 h 65 °C · 2–4 h 75 °C · 6–8 h 上限 ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h 上限 ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h 上限 ≤ 100 °C	60 °C · 3–4 h 80 °C · 6–8 h 90 °C · 6–8 h 上限 ≤ 100 °C

	耗材	PA6 GF 红色	PA6 CF 黑色	PA HT GF 红色	PA HT CF 黑色	PPA 黑色	PPA GF 黑色	PPA CF 黑色	PPA GF FR 黑色	PPS GF 灰色	PPS CF 黑色
特性	**强度** 弯曲强度 - XY 方向	140 MPa	162 MPa	153 MPa	172 MPa	123 MPa	159 MPa	256 MPa ▲	134 MPa	112 MPa	137 MPa
	强度 弯曲强度 - Z 方向	103 MPa ▲	91 MPa	102 MPa	95 MPa	62 MPa	75 MPa	80 MPa	62 MPa	58 MPa	62 MPa
	刚度 弯曲模量 - XY 方向	4375 MPa	5312 MPa	5152 MPa	6010 MPa	2672 MPa	5920 MPa	11384 MPa ▲	6346 MPa	4530 MPa	7402 MPa
	刚度 弯曲模量 - Z 方向	3054 MPa	2757 MPa	3142 MPa	3873 MPa	2576 MPa	3246 MPa	3278 MPa	4168 MPa ▲	2212 MPa	2885 MPa
	韧性 冲击强度 - XY 方向	63.4 kJ/m²	50.5 kJ/m²	35.5 kJ/m²	26.3 kJ/m²	32.8 kJ/m²	30.2 kJ/m²	34.2 kJ/m²	22.3 kJ/m²	31.7 kJ/m²	22.3 kJ/m²
	层间粘接 冲击强度 - Z 方向	15.8 kJ/m²	14.7 kJ/m²	10.5 kJ/m²	7.3 kJ/m²	5.4 kJ/m²	4.8 kJ/m²	4.6 kJ/m²	5.9 kJ/m²	10.6 kJ/m²	8.8 kJ/m²
	耐热性 热变形温度，0.45 MPa	191 °C	197 °C	195 °C	206 °C	103 °C（退火前）； 131 °C（退火后）	227 °C	227 °C	214 °C	241 °C	264 °C ▲
	长期耐热性 最高连续使用温度 （与负载、尺寸、形状、壁厚、含水率等有关）	100 - 120 °C（干态） （退火后的通常高于未退火的）	105 - 120 °C（干态） （退火后的通常高于未退火的）	105 - 120 °C（干态） （退火后的通常高于未退火的）	110 - 130 °C（干态） （退火后的通常高于未退火的）	100 - 125 °C（干态） （退火后的通常高于未退火的）	140 - 160 °C（干态） （退火后的通常高于未退火的）	145 - 160 °C（干态） （退火后的通常高于未退火的）	140 - 160 °C（干态） （退火后的通常高于未退火的）	190 - 210 °C （退火后的通常高于未退火的）	200 - 220 °C （退火后的通常高于未退火的） ▲
	饱和吸水率 25 °C，55% RH	2.33%	2.08%	1.03%	1.02%	1.32%	1.29%	1.30%	1.29%	0.05% ▲	0.05% ▲
	收缩率 (用特定模型与工艺测得，详见 TDS)	0.26% - 0.46%	0.17% - 0.44%	0.22% - 0.45%	0.18% - 0.43%	0.16% - 0.43%	0.14% - 0.39%	0.15% - 0.44%	0.17% - 0.43%	0.21% - 0.42%	0.19% - 0.41%
	表面硬度 邵氏硬度	82D	83D	84D	85D	87D	87D	88D	88D	83D	84D
	密度	1.22 g/cm³	1.15 g/cm³	1.23 g/cm³	1.15 g/cm³	1.14 g/cm³	1.28 g/cm	1.26 g/cm	1.32 g/cm³	1.36 g/cm	1.26 g/cm³
	熔融指数 (测试条件见 TDS)	17.8 g/10 min	14.7 g/10 min	17.8 g/10 min	12.6 g/10 min	15.6 g/10 min	13.2 g/10 min	12.4 g/10 min	17.8 g/10 min	22.3 g/10 min	20.6 g/10 min
前处理	使用前是否干燥										
	干燥条件	100–120 °C 8–14 h 鼓风干燥箱	100–120 °C 8–14 h 鼓风干燥箱	100–120 °C 8–14 h 鼓风干燥箱	100–120 °C 8–14 h 鼓风干燥箱	100–130 °C 10–14 h 鼓风干燥箱	100–140 °C 10–14 h 鼓风干燥箱	100–140 °C 10–14 h 鼓风干燥箱	100–140 °C 10–14 h 鼓风干燥箱	100–140 °C 6–8 h 鼓风干燥箱	100–140 °C 6–8 h 鼓风干燥箱
	是否兼容 AMS 或 CFS	✓	✓	✓	✓	✓	✗	✗	✗	✓	✓
	喷嘴尺寸/材料	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 常见喷嘴材质均可（黄铜、不锈钢、硬化钢等）	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质	0.2 0.4 0.6 0.8 硬化钢、碳化钨等加硬材质
	适用的打印底板和床温	光面 PEI / 纹理 PEI 80 – 100 °C	光面 PEI / 纹理 PEI 80 – 100 °C	光面 PEI / 纹理 PEI 90 – 120 °C	光面 PEI / 纹理 PEI 90 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C	光面 PEI / 纹理 PEI 100 – 120 °C
	粘接方式	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒	专用液体胶水 / 艾普创固体胶棒
打印设置	打印难度	●●●●○ 较难	●●●●○ 较难	●●●●○ 较难	●●●●○ 较难	●●●●● 很难	●●●●● 很难	●●●●● 很难	●●●●● 很难	●●●●● 很难	●●●●● 很难
	是否需要封箱打印	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	推荐 （封箱和适当高的腔温可让层间粘结更优）	必须 （不封箱和腔温偏低易导致层间粘接较弱）	必须 （不封箱和腔温偏低易导致层间粘接较弱）
	是否使用干燥剂密封防潮	必须	必须	必须	必须	必须	必须	必须	必须	必须	必须
	打印速度	最高达 300 mm/s	最高达 300 mm/s	最高达 250 mm/s	最高达 270 mm/s	最高达 180 mm/s	最高达 250 mm/s	最高达 250 mm/s	最高达 250 mm/s	最高达 300 mm/s	最高达 300 mm/s
	喷嘴温度	280 - 300 °C	280 - 300 °C	270 - 300 °C	270 - 300 °C	280 - 310 °C	280 - 310 °C	280 - 310 °C	280 - 300 °C	320 - 360 °C	320 - 360 °C
	部件冷却风扇	0% - 40%	0% - 40%	0% - 60%	0% - 60%	0% - 40%	0% - 40%	0% - 40%	0% - 40%	0% - 40%	0% - 40%
后处理	退火	60 °C · 3–4 h 90 °C · 6–8 h 上限 ≤ 120 °C（非必要不做）	60 °C · 3–4 h 90 °C · 6–8 h 上限 ≤ 120 °C（非必要不做）	70 °C · 2–4 h 90 °C · 6–8 h 上限 ≤ 120 °C（非必要不做）	70 °C · 2–4 h 90 °C · 6–8 h 上限 ≤ 120 °C（非必要不做）	70 °C · 2–4 h 90 °C · 6–8 h 上限 ≤ 120 °C（非必要不做）	60 °C · 2–4 h 90 °C · 2–4 h 120 °C · 6–8 h 上限 ≤ 160 °C	60 °C · 2–4 h 90 °C · 2–4 h 120 °C · 6–8 h 上限 ≤ 160 °C	60 °C · 2–4 h 90 °C · 2–4 h 120 °C · 6–8 h 上限 ≤ 150 °C	90 °C · 2–4 h 120 °C · 2–4 h 150 °C · 6–8 h 上限 ≤ 150 °C	90 °C · 2–4 h 120 °C · 2–4 h 150 °C · 6–8 h 上限 ≤ 180 °C

耗材应用 核心卖点 · 应用场景

TPU 95A HS 乳白

核心卖点

真**高速**（最高达 300 mm/s）
超**耐磨**（磨耗仅为 23 mm³）
多彩中国传统色；
易打印（适配入门级打印机）

应用场景

鞋类和包包、手机壳、玩具车轮胎
密封条和垫片、软质家居用品

TPU GF 70D 沧浪

核心卖点

玻纤纹理表面，**抗冲**耐摔、**耐磨**减震
经久耐用，抗冻**抗菌**，防水防油
极少拉丝，支持高速打印且易于上手
适配主流打印机，并兼容 AMS 或 CFS
多彩中国传统色（8 色）

应用场景

可用于打印玩具赛车及无人机配件
机器狗壳体、机械外罩、儿童足球
冰箱蛋架、游泳池管件、厨房沥水工具、柔性吧台垫、多色手机壳、展示品、装饰品及其他物品

TPU CF 70D 黑色

核心卖点

碳纤增强，**抗冲**耐摔、**耐磨**减震
经久耐用，抗冻抗菌，防水防油
极少拉丝，支持高速打印且易于上手
适配主流打印机，并兼容 AMS 或 CFS

应用场景

可用于打印蜂窝防撞、摩托车挡泥板、扬声器减震钉、柔性吧台垫及其他物品

PLA Matte 黝紫

核心卖点

30 种独家国潮色
优异**层间强度**和刚度
高级**哑光**质感
适配 AMS 或 CFS 狂暴模式打印

应用场景

玩具、潮玩手办、家居用品
文创摆件等外观件

PLA FC 哑蓝

核心卖点

大多数模型可免涂胶，食品可接触
安全环保，色彩丰富，高级**哑光**及细腻质感
高速易打印（最高速度 300 mm/s）
适配所有主流打印机、AMS 或 CFS

应用场景

厨房工具，食材容器，糕点模具
儿童玩具，宠物喂食器等
会与食品或口腔接触的物品

PLA CF 元青

核心卖点

碳纤增刚增强
超高**层间强度**
高级**磨砂**质感
多彩中国传统色
多种机型超速打印

应用场景

玩具、家居用品、文创礼品

PETG Metal 光刃银

核心卖点

强烈**金属光泽**，优雅高贵
超高韧性和**层间结合**强度
耐水耐油，耐候性升级，户外长期可用
易于打印，适配 AMS 或 CFS
支持高速（最高速度达 300 mm/s）

应用场景

潮玩手办、文创作品、艺术摆件
电子 DIY 产品外壳、文具盒
化妆品收纳盒、盆栽容器等

PETG GF 玄铁黑

核心卖点

柔和**金属光泽**，**磨砂**表面
强度、刚度与**层间结合**强度平衡
更少翘曲，更高精度耐水耐油
耐候性升级，户外长期可用；
易于打印，适配 AMS 或 CFS
支持高速（最高速度达 300 mm/s）

应用场景

潮玩手办、文创作品、艺术摆件
电子 DIY 产品外壳、电器外壳
置物钩架、地漏、盆栽容器等

PETG CF 玄铁银

核心卖点

高级金属观感，古色古香
更**高强度**和刚度
卓越韧性和**层间结合**强度
更少翘曲，更高精度；**耐水**耐油
耐候性升级，户外长期可用
易于打印，适配 AMS 或 CFS
支持高速（最高速度达 250 mm/s 以上）

应用场景

潮玩手办、文创作品、艺术摆件、电子 DIY 产品外壳、穿越机机架、遥控汽车壳体、盆栽容器等

PET DET & FC 蓝色

核心卖点

大多数模型可免涂胶，可探测
食品可接触，为食品、饮料、药品生产企业量身定制，**耐水耐油**，卫生安全，**高韧性**，易于打印、支持高速（最高速度 300 mm/s）
适配所有主流打印机、AMS 或 CFS

应用场景

食材罐、面粉勺、糕点模具等厨房工具，签字笔外壳、封口夹、抓取钳、配料盒等食品车间文具、工具，食品、饮品和药品加工设备零部件等会与此类产品接触的物品。

PP GF 灰色

核心卖点

超**低密度**，极致**轻量化**；
疏水，耐油，特耐化学腐蚀；
高韧性，高层间结合强度，耐疲劳，耐弯折；
较宽使用温度范围（约 -20 °C ~ 100 °C）
玻璃纤维增强，表面磨砂，绝缘性优；
更少翘曲，更易打印，适配主流打印机
适配 AMS、CFS 等多色供料系统
多种颜色可选

应用场景

要求轻量化的车载支架和玩具赛车、无人机、机器人配件，要求耐酸碱、溶剂、油脂的药剂取用工具、电镀夹具、清洗台架，要求耐疲劳、耐弯折的卡扣、铰链和活动关节，要求绝缘的电子元器件外壳

PP CF 黑色

核心卖点

超**低密度**，极致**轻量化**；
疏水，耐油，特耐化学腐蚀；
高韧性，高层间结合强度，耐疲劳，耐弯折；较宽使用温度范围（约 -20 °C ~ 100 °C）
碳纤维增强，表面平滑细腻；
更少翘曲，更易打印，适配主流打印机，
适配 AMS、CFS 等多色供料系统

应用场景

要求轻量化的车载支架和玩具赛车、无人机、机器人配件，要求耐酸碱、溶剂、油脂的药剂取用工具、电镀夹具、清洗台架，要求耐疲劳、耐弯折的卡扣、铰链和活动关节

PET GF 蓝色

核心卖点

高强度，高刚度，更**高韧性**和**层间结合**强度
承重抗压；低翘曲，耐蠕变
更高尺寸精度；耐水耐油，耐候性升级
室内水环境和户外长期可用；易于打印，
适配 AMS 或 CFS，支持高速
（最高速度达 250 mm/s 以上）
多种颜色可选

应用场景

功能原型，工装夹具，电动工具零部件，电子电器外壳，高压水枪外壳，水管连接件，工具箱把手等

PET CF 黑色

核心卖点

高强度，高刚度，更**高韧性**和**层间结合**强度，承重抗压；低翘曲，耐蠕变，更高尺寸精度；耐水耐油，耐候性升级，室内水环境和户外长期可用；易于打印，适配 AMS 或 CFS，支持高速（最高速度达 300 mm/s 以上）

应用场景

功能原型，工装夹具，电动工具零部件，电子电器外壳，高压水枪外壳，水管连接件，工具箱把手等

ASA GF 磨砂蓝

核心卖点

极高的**抗紫外线**和**耐候性**，出色的**耐水**性和耐热性，低密度，高刚性，超低翘曲（无需主动加热舱室），优异的尺寸稳定性和表面质量，兼容 AMS 或 CFS，支持超高速打印
（最高可达 300 mm/s），且支持多色打印

应用场景

园艺花盆、露营工具箱、露台灯、鸟屋、游泳池管件、天气监测设备外壳、船舶零件、飞机模型、船模，以及其他用于户外及涉水环境，或需要轻量化与高刚性的物品。

ASA CF 黑色

核心卖点

顶级**耐候**，较好的**耐热性**，**低密度**，低吸水，强度与抗冲性能双优，超低翘曲（无需腔体加热），优异的尺寸稳定性和表面质量，兼容AMS 或 CFS，支持超高速打印
（最高可达300 mm/s）

应用场景

鸟屋、船舶零件、育苗盘、飞机模型、船模，以及其他水环境应用的户外功能件以及船模和航模等轻量化部件。

PC FR HS 苍苍色

核心卖点

无氟无卤，超薄**阻燃**（0.8 mm V-0）
国潮美学，半透高韧
优异层间，**超低翘曲**
超速打印（最高达 350 mm/s）

应用场景

智能灯具、移动充电宝、网络配线架和电脑硬盘笼外壳等要求顶级阻燃且无氟无卤的消费电子产品和户外炉三角架等高强耐热部件

PC GF 石榴裙

核心卖点

超低翘曲（无需腔体加热）
超低吸水、**耐候**升级
多彩中国传统色和优异表面质量
适配 AMS 或 CFS 超速打印

应用场景

冲浪板和水泵等水环境应用部件、电源外壳等户外应用的高刚高强绝缘部件

PC CF 黑色

核心卖点

超低翘曲（无需腔体加热）
超低吸水、**耐候**升级
超高层间性能和优异表面质量
超速打印（最高达 350 mm/s）

应用场景

冲浪板和水泵等水环境应用部件、户外应用的高刚高强部件，齿轮等精密耐磨部件

CoPA HS 红色

核心卖点

表面光洁平滑，低摩擦系数，**耐磨**，**耐油**、耐常规化学试剂，干态**高强度**、高刚性、高韧性，湿态超高韧性，超低翘曲，适配 AMS 或 CFS 打印，黑、白、灰、红 4 大基础色系

应用场景

齿轮、滑轨、轴套等要求耐磨的部件，机械臂、无人机支架、玩具赛车、摩托车挡泥板等要求强度、刚度与韧性平衡的部件

PA6 GF 红色

核心卖点

玻纤增强，刚强**耐热**，不易翘曲
卓越层间，高韧耐冲，**绝缘**性优
黑、白、灰、红四大基础色系
支持 AMS 或 CFS 多色打印

应用场景

RC 赛车、无人机零部件、电动工具、吹风机喷嘴等轻质、高强、高韧、高绝缘、高耐热需求的应用

PA6 CF 黑色

核心卖点

碳纤增强，刚强**耐热**，不易翘曲
卓越层间，高韧耐冲
适配 AMS 或 CFS 打印

应用场景

烘箱托盘和架子、汽车、无人机、RC 赛车、电动工具等高耐热、轻质、高强、高韧需求的应用

PA HT GF 红色

核心卖点

玻纤增强，**超低翘曲**，高精**耐磨**
卓越层间，刚强抗冲，优异耐热性
黑、白、灰、红四大基础色系
支持 AMS 或 CFS 多色打印

应用场景

汽车、电动工具等轻质、绝缘、高性能需求的应用

PA HT CF 黑色

核心卖点

碳纤增强，**超低翘曲**，高精**耐磨**
卓越层间，刚强抗冲，优异耐热性
适配 AMS 或 CFS 打印

应用场景

汽车、无人机、RC赛车、电动工具等轻质、高性能需求的应用

PPA 黑色

核心卖点

高强、高韧、极致耐温耐湿
超低翘曲、**抗蠕变**、高表面精度
黑、白、灰、红四大基础色系
适配 AMS 或 CFS 打印

应用场景

自行车配件、电子设备和智能家居的耐温耐湿绝缘部件

PPA GF 黑色

核心卖点

刚韧均衡、极致耐温耐湿
超低收缩和翘曲、**绝缘**
黑、白、灰、红四大基础色系

应用场景

电子设备和智能家居的绝缘结构件、汽车和机械行业金属部件取代

PPA CF 黑色

核心卖点

碳纤刚强、金属取代
超低吸湿、超高耐温
高精**耐磨**、耐蠕变
超低翘曲、**高速**易打

应用场景

无人机、机器人、汽车和机械行业金属部件取代

PPA GF FR 黑色

核心卖点

顶级**阻燃**、**绝缘耐热**
高刚低翘、超低吸湿
无需腔体加热、广泛兼容多种机型
黑、白、橙、蓝经典四色

应用场景

阻燃接线盒、插座部件、绝缘固定件等
电子电气原型与零件；生产线装配夹具、
检测工装等耐热工具与治具
办公设备和智能硬件等高要求零件

PPS GF 灰色

核心卖点

玻纤增强、顶级**阻燃**、**耐热**绝缘
高刚高强、耐水抗化
层间优化、广泛兼容高温型打印机
适配 AMS 或 CFS 易打印

应用场景

电机支架、电动滑板转动齿轮等
高强度工业部件；汽车涡轮进气管、
新能源汽车电池支架、高温电气设备壳体等极致耐高温绝缘部件
手术刀柄、化工泵等耐水抗化部件

PPS CF 黑色

核心卖点

碳纤刚强、顶级**阻燃**
极致**耐水耐高温**耐溶剂
层间优化、广泛兼容高温型打印机
适配 AMS 或 CFS 易打印

应用场景

电机冷却风扇、汽车加热器转子、电动滑板齿轮、数控机床电机轴座等高强度工业部件；涡轮进气管等阻燃部件
摩托车邮箱等耐水耐高温部件

数据版本 2026.10.14 · 数据来源：ipconpolymer.com · 具体参数以官方 TDS 为准 · 点击产品名查看官网 · 第 1 / 2 页
难度为参考评级，带 * 为按家族推算；喷嘴 chip：黑色=支持、蓝色=推荐；底色越深=本行相对水平，▲/▼=最优（硬度/熔指不标最优）；层间粘接条按数值推导。

条长 = 行内相对水平（按数值）；颜色 = 行主题色；细条 = 行内相对水平　TDS 下载 ↗

IPCON Filament Guide

Consumables Selection

This filament guide provides a comprehensive comparison of the performance, printing requirements and applications of IPCON consumables, helping users select materials that best match their needs.

For detailed technical specifications, download the Technical Data Sheet (TDS) on the product page.

Contact us
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www.ipconpolymer.com ↗

Bar length = relative level (by value) Theme: ● Strength ● Stiffness ● Toughness/Layer ● Thermal ● Moisture ● Hardness ● Lightweight ● Flow ●●●○○ Difficulty Required Recommended Optional Temp / Speed / Fan range Tint = row theme, darker = better Thin bar = relative level (longer = better)

	Consumables	TPU 95A HS Milky White	TPU GF 70D Mountain Blue	TPU CF 70D Black	PLA Matte Can Zi	PLA FC Matte Blue	PLA CF Dark Cyan	PETG Metal Blade Silver	PETG GF Forged Iron Black	PETG CF Forged Iron Silver	PET DET & FC Blue
Filament Properties	● Strength Flexural Strength - XY	27 MPa (Tensile)	14 MPa (no break)	22 MPa (no break)	67 MPa	64 MPa	89 MPa	65 MPa	78 MPa	84 MPa	63 MPa
	Layer Strength Flexural Strength - Z	22 MPa (Tensile)	12 MPa (no break)	14 MPa (no break)	42 MPa	40 MPa	68 MPa	59 MPa	53 MPa	57 MPa	41 MPa
	● Stiffness Flexural Modulus - XY	—	280 MPa	427 MPa	3330 MPa	2933 MPa	4429 MPa	1955 MPa	2934 MPa	3488 MPa	1752 MPa
	Layer Stiffness Flexural Modulus - Z	—	235 MPa	261 MPa	2759 MPa	2693 MPa	2663 MPa	1841 MPa	2132 MPa	2376 MPa	1667 MPa
	● Toughness Impact Strength - XY	> 125.0 kJ/m² (no break) ▲	> 124.0 kJ/m² (no break)	> 124.0 kJ/m² (no break)	23.8 kJ/m²	29.6 kJ/m²	21.9 kJ/m²	76.8 kJ/m²	21.1 kJ/m²	37.1 kJ/m²	47.8 kJ/m²
	● Layer Adhesion Impact Strength - Z	> 125.0 kJ/m² ▲	18.2 kJ/m²	41.4 kJ/m²	5.2 kJ/m²	6.1 kJ/m²	9.0 kJ/m²	20.8 kJ/m²	10.4 kJ/m²	12.9 kJ/m²	7.3 kJ/m²
	● Heat Resistance HDT, 0.45 MPa	—	—	—	56 °C	55 °C	55.4 °C	68 °C	74 °C	74 °C	71 °C
	● Long-term Heat Resistance Relative Thermal Index (RTI) Max Continuous Service Temperature (Read the TDS for more details)	70 - 80 °C	70 - 90 °C	70 - 90 °C	45 - 50 °C (The value after annealing is higher than that before annealing in general)	45 - 50 °C (The value after annealing is higher than that before annealing in general)	45 - 53 °C (The value after annealing is higher than that before annealing in general)	60 - 65 °C	60 - 68 °C	60 - 68 °C	60 - 65 °C
	● Saturated Water Absorption Rate 25 °C, 55% RH	1.04%	1.13%	1.16%	0.43%	0.40%	0.42%	0.34%	0.31%	0.32%	0.37%
	● Shrinkage Rate (Tested with specific models and parameters; read the TDS for more details)	0.18% - 0.37%	0.26% - 0.45%	0.13% - 0.42%	0.23% - 0.41%	0.03% - 0.30% ▲	0.13% - 0.38%	0.26% - 0.51%	0.17% - 0.37%	0.08% - 0.41%	0.28% - 0.51%
	● Surface Hardness Shore Hardness	95A	~ 70D	~ 70D	80D	81D	81D	75D	77D	78D	78D
	● Density	1.21 g/cm³	1.19 g/cm³	1.09 g/cm³	1.39 g/cm³	1.39 g/cm³	1.28 g/cm³	1.26 g/cm³	1.32 g/cm³	1.28 g/cm³	1.39 g/cm³
	● Melting Index (Read the TDS for more details)	28.4 g/10 min	10.2 g/10 min	3.5 g/10 min	17.4 g/10 min	12.0 g/10 min	8.0 g/10 min	25.8 g/10 min	19.0 g/10 min	17.5 g/10 min	25.9 g/10 min
Pre Processing	Dry Out Before Use	All products After unpacking, use directly with desiccants inside a sealed container. If it becomes damp, re-drying is required									
	Drying Condition	70–90 °C 8–12 h Blast Drying Oven	80–100 °C 8–12 h Blast Drying Oven	80–100 °C 8–12 h Blast Drying Oven	55–60 °C 6–8 h Blast Drying Oven	55–60 °C 6–8 h Blast Drying Oven	55–70 °C 8–12 h Blast Drying Oven	60–65 °C 6–8 h Blast Drying Oven	65–70 °C 6–8 h Blast Drying Oven	65–70 °C 6–8 h Blast Drying Oven	65–70 °C 6–8 h Blast Drying Oven
	AMS and CFS Compatibility	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Nozzle Size / Material	0.2 0.4 0.6 0.8 All Material	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide and other hardened materials	0.2 0.4 0.6 0.8 All Material	0.2 0.4 0.6 0.8 Stainless steel, hardened steel, tungsten carbide, diamond and other materials with very low risk of containing heavy metals (consult the nozzle manufacturer for details)	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 All Material	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Stainless steel, hardened steel, tungsten carbide, and other materials with very low risk of containing heavy metals (consult the nozzle manufacturer for details)
	Build Plate & Bed Temperature	Smooth / Textured PEI Plate 30 – 40 °C	Smooth / Textured PEI Plate / Textured PEI Plate 35 – 50 °C	Smooth / Textured PEI Plate / Textured PEI Plate 35 – 50 °C	Smooth / Textured PEI Plate 35 – 65 °C	Smooth / Textured PEI Plate / Textured PEI Plate 35 – 65 °C	Smooth / Textured PEI Plate 30 – 50 °C	Smooth / Textured PEI Plate / Textured PEI Plate 65 – 75 °C	Smooth / Textured PEI Plate / Textured PEI Plate 65 – 75 °C	Smooth / Textured PEI Plate / Textured PEI Plate 65 – 75 °C	Smooth / Textured PEI Plate / Textured PEI Plate 65 – 80 °C
	Adhesion Methods	Apply liquid glue or IPCON solid glue stick (PVP) if the first layer bonding is not firm enough	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Apply liquid glue or IPCON solid glue stick (PVP) if the first layer bonding is not firm enough	Food contact use: do not use glue or apply food grade glue Non-food contact use: apply liquid glue or PVP solid glue stick	Apply liquid glue or IPCON solid glue stick (PVP) if the first layer bonding is not firm enough	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Food contact use: do not use glue or apply food grade glue Non-food contact use: apply liquid glue or PVP solid glue stick
Printer Settings	Printing Difficulty	●●●○○ Moderate	●●●○○ Moderate	●●●○○ Moderate	●○○○○ Easy	●○○○○ Easy	●●○○○ Fairly easy	●●○○○ Fairly easy	●●○○○ Fairly easy	●●○○○ Fairly easy	●●●○○ Moderate
	Print with Enclosure	Optional	Optional	Optional	Optional	Optional	Optional	Optional	Optional	Optional	Optional
	Seal with Desiccant and Protect from Moisture	Required	Required	Required	Recommended (Better results will get when it is very dry and sealed)	Recommended (Better results will get when it is very dry and sealed)	Recommended (Better results will get when it is very dry and sealed)	Required	Required	Required	Required
	Print Speed	Up to 300 mm/s	Up to 250 mm/s	Up to 250 mm/s	Up to 300 mm/s	Up to 300 mm/s	Up to 300 mm/s	Up to 300 mm/s	Up to 300 mm/s	Up to 250 mm/s	Up to 300 mm/s
	Nozzle Temperature	220 - 240 °C	210 - 250 °C	210 - 250 °C	210 - 240 °C	190 - 230 °C	210 - 240 °C	240 - 270 °C	245 - 270 °C	245 - 270 °C	270 - 300 °C
	Part Cooling Fan	80% - 100%	10% - 90%	10% - 90%	60% - 100%	60% - 100%	60% - 100%	10% - 90%	10% - 90%	10% - 90%	10% - 80%
Post Processing	Annealing	No annealing required If needed: ≤ 80 °C · 6 h	No annealing required If needed: ≤ 100 °C · 6 h	No annealing required If needed: ≤ 100 °C · 6 h	45 °C · 2–4 h 50 °C · 2–4 h 55 °C · 4–6 h max ≤ 60 °C	45 °C · 2–4 h 50 °C · 2–4 h 55 °C · 4–8 h max ≤ 60 °C	45 °C · 2–4 h 50 °C · 2–4 h 55 °C · 4–6 h max ≤ 60 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h max ≤ 70 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h max ≤ 70 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h max ≤ 70 °C	55 °C · 2–4 h 60 °C · 2–4 h 65 °C · 4–8 h max ≤ 75 °C

	Consumables	PP GF Grey	PP CF Black	PET GF Blue	PET CF Black	ASA GF Matte Blue	ASA CF Black	PC FR HS Navy Blue	PC GF Pomegranate Red	PC CF Black	CoPA HS Red
Filament Properties	● Strength Flexural Strength - XY	53 MPa	46 MPa	113 MPa	134 MPa	76 MPa	80 MPa	101 MPa	108 MPa	116 MPa	113 MPa
	Layer Strength Flexural Strength - Z	31 MPa	30 MPa	75 MPa	62 MPa	49 MPa	45 MPa	75 MPa	62 MPa	73 MPa	65 MPa
	● Stiffness Flexural Modulus - XY	1464 MPa	1237 MPa	4067 MPa	5162 MPa	2758 MPa	3171 MPa	2521 MPa	3367 MPa	5482 MPa	3068 MPa
	Layer Stiffness Flexural Modulus - Z	664 MPa	629 MPa	2511 MPa	3136 MPa	2156 MPa	2173 MPa	2754 MPa	1948 MPa	2362 MPa	2385 MPa
	● Toughness Impact Strength - XY	46.5 kJ/m²	56.8 kJ/m²	40.7 kJ/m²	22.5 kJ/m²	23.5 kJ/m²	25.7 kJ/m²	62.6 kJ/m²	34.2 kJ/m²	23.7 kJ/m²	64.3 kJ/m²
	● Layer Adhesion Impact Strength - Z	13.4 kJ/m²	23.3 kJ/m²	15.4 kJ/m²	9.1 kJ/m²	8.2 kJ/m²	7.7 kJ/m²	11.2 kJ/m²	12.4 kJ/m²	16.1 kJ/m²	13.3 kJ/m²
	● Heat Resistance HDT, 0.45 MPa	0.45 MPa unannealed: 98 °C 0.45 MPa annealed: 104 °C 1.80 MPa unannealed: 84 °C 1.80 MPa annealed: 89 °C	0.45 MPa unannealed: 113 °C 0.45 MPa annealed: 115 °C 1.80 MPa unannealed: 81 °C 1.80 MPa annealed: 87 °C	0.45 MPa unannealed: 74 °C 0.45 MPa annealed: 99 °C 1.80 MPa unannealed: 66 °C 1.80 MPa annealed: 73 °C	0.45 MPa unannealed: 138 °C 0.45 MPa annealed: 155 °C 1.80 MPa unannealed: 76 °C 1.80 MPa annealed: 87 °C	98 °C	99 °C	92 °C	109 °C	113 °C	0.45 MPa unannealed: 150 °C 0.45 MPa annealed: 152 °C 1.80 MPa unannealed: 65 °C 1.80 MPa annealed: 70 °C
	● Long-term Heat Resistance Relative Thermal Index (RTI) Max Continuous Service Temperature (Read the TDS for more details)	75 - 85 °C (The value after annealing is higher than that before annealing in general)	75 - 85 °C (The value after annealing is higher than that before annealing in general)	65 - 75 °C (The value after annealing is higher than that before annealing in general)	65 - 80 °C (The value after annealing is higher than that before annealing in general)	75 - 85 °C	75 - 85 °C	70 - 85 °C	85 - 100 °C	85 - 100 °C	60 - 80 °C (Dry State) (The value after annealing is higher than that before annealing in general)
	● Saturated Water Absorption Rate 25 °C, 55% RH	< 0.06%	< 0.06%	0.36%	0.38%	0.31%	0.33%	0.28%	0.22%	0.21%	2.98%
	● Shrinkage Rate (Tested with specific models and parameters; read the TDS for more details)	0.25% - 0.57%	0.19% - 0.59%	0.23% - 0.43%	0.17% - 0.46%	0.13% - 0.37%	0.14% - 0.32%	0.39% - 0.50%	0.29% - 0.47%	0.21% - 0.44%	0.28% - 0.49%
	● Surface Hardness Shore Hardness	70D	70D	85D	86D	80D	81D	84D	85D	85D	84D
	● Density	1.04 g/cm³	0.92 g/cm³ ▲	1.36 g/cm³	1.26 g/cm³	1.11 g/cm³	1.04 g/cm³	1.22 g/cm³	1.30 g/cm³	1.24 g/cm³	1.14 g/cm³
	● Melting Index (Read the TDS for more details)	11.9 g/10 min	10.8 g/10 min	15.4 g/10 min	17.8 g/10 min	19.5 g/10 min	17.7 g/10 min	19.4 g/10 min	16.4 g/10 min	14.7 g/10 min	16.3 g/10 min
Pre Processing	Dry Out Before Use										
	Drying Condition	70–100 °C 6–8 h Blast Drying Oven	70–100 °C 6–8 h Blast Drying Oven	80–110 °C 6–10 h Blast Drying Oven	80–110 °C 6–10 h Blast Drying Oven	70–90 °C 4–8 h Blast Drying Oven	70–90 °C 4–8 h Blast Drying Oven	65–75 °C 8–12 h Blast Drying Oven	80–100 °C 8–12 h Blast Drying Oven	80–100 °C 8–12 h Blast Drying Oven	100–120 °C 8–14 h Blast Drying Oven
	AMS and CFS Compatibility	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Nozzle Size / Material	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 All Material	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 All Material
	Build Plate & Bed Temperature	Smooth / Textured PEI Plate / Textured PEI Plate 55 – 75 °C	Smooth / Textured PEI Plate / Textured PEI Plate 55 – 75 °C	Smooth / Textured PEI Plate / Textured PEI Plate 70 – 90 °C	Smooth / Textured PEI Plate / Textured PEI Plate 70 – 90 °C	Smooth / Textured PEI Plate / Textured PEI Plate 90 – 110 °C	Smooth / Textured PEI Plate / Textured PEI Plate 90 – 110 °C	Smooth / Textured PEI Plate / Textured PEI Plate 85 – 100 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 90 – 110 °C
	Adhesion Methods	Special Glue, film or plate for PP	Special Glue, film or plate for PP	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	IPCON solid glue stick (PVP)	IPCON solid glue stick (PVP)	IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)
Printer Settings	Printing Difficulty	●●●●○ Challenging*	●●●●○ Challenging*	●●●○○ Moderate*	●●●○○ Moderate*	●●●○○ Moderate	●●●○○ Moderate	●●●●○ Challenging	●●●●○ Challenging	●●●●○ Challenging	●●●●○ Challenging*
	Print with Enclosure	Optional	Optional	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)
	Seal with Desiccant and Protect from Moisture	Required	Required	Required	Required	Required	Required	Required	Required	Required	Required
	Print Speed	Up to 250 mm/s	Up to 220 mm/s	Up to 250 mm/s	Up to 300 mm/s	Up to 300 mm/s	Up to 300 mm/s	Up to 350 mm/s	Up to 350 mm/s	Up to 350 mm/s	Up to 300 mm/s
	Nozzle Temperature	220 - 260 °C	220 - 260 °C	265 - 295 °C	265 - 295 °C	265 - 290 °C	265 - 290 °C	240 - 270 °C	280 - 300 °C	280 - 300 °C	270 - 300 °C
	Part Cooling Fan	10% - 70%	10% - 70%	10% - 40%	10% - 40%	10% - 40%	10% - 40%	0% - 60%	0% - 40%	0% - 40%	0% - 40%
Post Processing	Annealing	60 °C · 3–4 h 70 °C · 3–4 h 80 °C · 6–8 h max ≤ 90 °C	60 °C · 3–4 h 70 °C · 3–4 h 80 °C · 6–8 h max ≤ 90 °C	60 °C · 3–4 h 80 °C · 3–4 h 90 °C · 6–8 h max ≤ 100 °C	60 °C · 3–4 h 80 °C · 3–4 h 90 °C · 6–8 h max ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h max ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h max ≤ 100 °C	55 °C · 2–4 h 65 °C · 2–4 h 75 °C · 6–8 h max ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h max ≤ 100 °C	65 °C · 2–4 h 75 °C · 2–4 h 80 °C · 4–8 h max ≤ 100 °C	60 °C · 3–4 h 80 °C · 6–8 h 90 °C · 6–8 h max ≤ 100 °C

	Consumables	PA6 GF Red	PA6 CF Black	PA HT GF Red	PA HT CF Black	PPA Black	PPA GF Black	PPA CF Black	PPA GF FR Black	PPS GF Grey	PPS CF Black
Filament Properties	● Strength Flexural Strength - XY	140 MPa	162 MPa	153 MPa	172 MPa	123 MPa	159 MPa	256 MPa ▲	134 MPa	112 MPa	137 MPa
	Layer Strength Flexural Strength - Z	103 MPa	91 MPa	102 MPa	95 MPa	62 MPa	75 MPa	80 MPa	62 MPa	58 MPa	62 MPa
	● Stiffness Flexural Modulus - XY	4375 MPa	5312 MPa	5152 MPa	6010 MPa	2672 MPa	5920 MPa	11384 MPa ▲	6346 MPa	4530 MPa	7402 MPa
	Layer Stiffness Flexural Modulus - Z	3054 MPa	2757 MPa	3142 MPa	3873 MPa	2576 MPa	3246 MPa	3278 MPa	4168 MPa	2212 MPa	2885 MPa
	● Toughness Impact Strength - XY	63.4 kJ/m²	50.5 kJ/m²	35.5 kJ/m²	26.3 kJ/m²	32.8 kJ/m²	30.2 kJ/m²	34.2 kJ/m²	22.3 kJ/m²	31.7 kJ/m²	22.3 kJ/m²
	● Layer Adhesion Impact Strength - Z	15.8 kJ/m²	14.7 kJ/m²	10.5 kJ/m²	7.3 kJ/m²	5.4 kJ/m²	4.8 kJ/m²	4.6 kJ/m²	5.9 kJ/m²	10.6 kJ/m²	8.8 kJ/m²
	● Heat Resistance HDT, 0.45 MPa	191 °C	197 °C	195 °C	206 °C	103 °C (before annealing); 131 °C (after annealing)	227 °C	227 °C	214 °C	241 °C	264 °C ▲
	● Long-term Heat Resistance Relative Thermal Index (RTI) Max Continuous Service Temperature (Read the TDS for more details)	100 - 120 °C (Dry State) (The value after annealing is higher than that before annealing in general)	105 - 120 °C (Dry State) (The value after annealing is higher than that before annealing in general)	105 - 120 °C (Dry State) (The value after annealing is higher than that before annealing in general)	110 - 130 °C (Dry State) (The value after annealing is higher than that before annealing in general)	100 - 125 °C (Dry State) (The value after annealing is higher than that before annealing in general)	140 - 160 °C (Dry State) (The value after annealing is higher than that before annealing in general)	145 - 160 °C (Dry State) (The value after annealing is higher than that before annealing in general)	140 - 160 °C (Dry State) (The value after annealing is higher than that before annealing in general)	190 - 210 °C (The value after annealing is higher than that before annealing in general)	200 - 220 °C (The value after annealing is higher than that before annealing in general) ▲
	● Saturated Water Absorption Rate 25 °C, 55% RH	2.33%	2.08%	1.03%	1.02%	1.32%	1.29%	1.30%	1.29%	0.05% ▲	0.05% ▲
	● Shrinkage Rate (Tested with specific models and parameters; read the TDS for more details)	0.26% - 0.46%	0.17% - 0.44%	0.22% - 0.45%	0.18% - 0.43%	0.16% - 0.43%	0.14% - 0.39%	0.15% - 0.44%	0.17% - 0.43%	0.21% - 0.42%	0.19% - 0.41%
	● Surface Hardness Shore Hardness	82D	83D	84D	85D	87D	87D	88D	88D	83D	84D
	● Density	1.22 g/cm³	1.15 g/cm³	1.23 g/cm³	1.15 g/cm³	1.14 g/cm³	1.28 g/cm³	1.26 g/cm³	1.32 g/cm³	1.36 g/cm³	1.26 g/cm³
	● Melting Index (Read the TDS for more details)	17.8 g/10 min	14.7 g/10 min	17.8 g/10 min	12.6 g/10 min	15.6 g/10 min	13.2 g/10 min	12.4 g/10 min	17.8 g/10 min	22.3 g/10 min	20.6 g/10 min
Pre Processing	Dry Out Before Use										
	Drying Condition	100–120 °C 8–14 h Blast Drying Oven	100–120 °C 8–14 h Blast Drying Oven	100–120 °C 8–14 h Blast Drying Oven	100–120 °C 8–14 h Blast Drying Oven	100–130 °C 10–14 h Blast Drying Oven	100–140 °C 10–14 h Blast Drying Oven	100–140 °C 10–14 h Blast Drying Oven	100–140 °C 10–14 h Blast Drying Oven	100–140 °C 6–8 h Blast Drying Oven	100–140 °C 6–8 h Blast Drying Oven
	AMS and CFS Compatibility	✓	✓	✓	✓	✓	✗	✗	✗	✓	✓
	Nozzle Size / Material	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Common nozzle materials (brass, stainless steel, hardened steel, etc.)	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials	0.2 0.4 0.6 0.8 Hardened steel, tungsten carbide, and other hardened materials
	Build Plate & Bed Temperature	Smooth / Textured PEI Plate / Textured PEI Plate 80 – 100 °C	Smooth / Textured PEI Plate / Textured PEI Plate 80 – 100 °C	Smooth / Textured PEI Plate 90 – 120 °C	Smooth / Textured PEI Plate 90 – 120 °C	Smooth / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C	Smooth / Textured PEI Plate / Textured PEI Plate 100 – 120 °C
	Adhesion Methods	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)	Liquid glue or IPCON solid glue stick (PVP)
Printer Settings	Printing Difficulty	●●●●○ Challenging	●●●●○ Challenging	●●●●○ Challenging	●●●●○ Challenging	●●●●● Very hard	●●●●● Very hard	●●●●● Very hard	●●●●● Very hard	●●●●● Very hard	●●●●● Very hard
	Print with Enclosure	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Recommended (Better layer adhesion will get with relatively higher chamber temperature)	Required (Worse layer adhesion may get with low chamber temperature)	Required (Worse layer adhesion may get with low chamber temperature)
	Seal with Desiccant and Protect from Moisture	Required	Required	Required	Required	Required	Required	Required	Required	Required	Required
	Print Speed	Up to 300 mm/s	Up to 300 mm/s	Up to 250 mm/s	Up to 270 mm/s	Up to 180 mm/s	Up to 250 mm/s	Up to 250 mm/s	Up to 250 mm/s	Up to 300 mm/s	Up to 300 mm/s
	Nozzle Temperature	280 - 300 °C	280 - 300 °C	270 - 300 °C	270 - 300 °C	280 - 310 °C	280 - 310 °C	280 - 310 °C	280 - 300 °C	320 - 360 °C	320 - 360 °C
	Part Cooling Fan	0 - 40%	0 - 40%	0 - 60%	0 - 60%	0% - 40%	0% - 40%	0% - 40%	0% - 40%	0% - 40%	0% - 40%
Post Processing	Annealing	60 °C · 3–4 h 90 °C · 6–8 h max ≤ 120 °C (avoid unless necessary)	60 °C · 3–4 h 90 °C · 6–8 h max ≤ 120 °C (avoid unless necessary)	70 °C · 2–4 h 90 °C · 6–8 h max ≤ 120 °C (avoid unless necessary)	70 °C · 2–4 h 90 °C · 6–8 h max ≤ 120 °C (avoid unless necessary)	70 °C · 2–4 h 90 °C · 6–8 h max ≤ 120 °C (avoid unless necessary)	60 °C · 2–4 h 90 °C · 2–4 h 120 °C · 6–8 h max ≤ 160 °C	60 °C · 2–4 h 90 °C · 2–4 h 120 °C · 6–8 h max ≤ 160 °C	60 °C · 2–4 h 90 °C · 2–4 h 120 °C · 6–8 h max ≤ 150 °C	90 °C · 2–4 h 120 °C · 2–4 h 150 °C · 6–8 h max ≤ 150 °C	90 °C · 2–4 h 120 °C · 2–4 h 150 °C · 6–8 h max ≤ 180 °C

Filament USP & applications

USP · Applications

TPU 95A HS Milky White

USP

Ultra-high speed printing (up to 300 mm/s);
Excellent **wear resistance** (only 23 mm³);
Rich traditional Chinese colors;
Easy to print
(compatible with most printers)

Applications

Shoes and bags,Phone case,Toy car tire,Soft home furnishings,Industrial Seal/gasket/ washer/belt

TPU GF 70D Mountain Blue

USP

Glass fiber-textured surface, impact and drop resistant, wear-resistant and shock-absorbing, durable, freeze-resistant and antibacterial, water and oil resistant, minimal stringing, supports high-speed printing and is easy to use, compatible with mainstream printers, AMS/CFS. Rich colors (8 traditional Chinese colors)

Applications

Can be used to print toy racing cars and drone accessories, robot dog shells, mechanical skins, children's footballs, refrigerator egg holders, swimming pool pipe fittings, kitchen draining tools, flexible bar mats, multi-color mobile phone cases, displays, decorations, and other items

TPU CF 70D Black

USP

Carbon fiber reinforced, impact and drop resistant, wear-resistant and shock-absorbing, durable, freeze-resistant and antibacterial, water and oil resistant, minimal stringing, supports high-speed printing and is easy to use, compatible with mainstream printers, AMS/CFS

Applications

Can be used to print honeycomb impact protection, motorcycle fenders, speaker damping spikes, flexible bar mats, and other items

PLA Matte Can Zi

USP

30 unique traditional Chinese colors and **matte** texture;
High stiffness and layer strength;
Can run ludicrous model with AMS and CFS

Applications

Toys and figures, Household articles, Cultural and creative gifts

PLA FC Matte Blue

USP

Most models are glue-free, **food-contact** safe, and environmentally friendly. Offer a wide range of vibrant colors with a premium matte finish and smooth texture. Designed for high-speed, easy printing (up to 300 mm/s), compatible with all major printers, AMS/CFS

Applications

Kitchen tools, food containers, pastry molds, children's toys, pet feeders, and other items that come into contact with food or the mouth

PLA CF Dark Cyan

USP

Carbon fiber reinforced;
Excellent stiffness and layer strength;
Unique **matte** texture and rich traditional Chinese colors;
Ultra-high speed printing with AMS and CFS

Applications

Toys and figures, Household articles, Cultural and creative gifts

PETG Metal Blade Silver

USP

Strong **metallic** luster, elegant and noble;
Ultra-high toughness and interlayer adhesion;
Water and oil resistance, upgraded weather resistance for long-term outdoor use;
AMS/CFS compatibility for easy and ultra-fast printing (up to 300 mm/s)

Applications

Designer toys, cultural and creative products, artistic ornaments, electronic DIY product enclosures, pencil cases, cosmetic storage boxes, planter pots, etc.

PETG GF Forged Iron Black

USP

Soft **metallic** luster with a matte surface;
Balanced strength, stiffness, and interlayer adhesion;
Low warping and higher precision;
Water and oil resistance, upgraded weather resistance for long-term outdoor use;
AMS/CFS compatibility for easy and ultra-fast printing
(up to 300 mm/s)

Applications

Designer toys, cultural and creative products, artistic ornaments, electronic DIY product enclosures, appliance housings, storage hooks and racks, floor drains, planter pots, etc.

PETG CF Forged Iron Silver

USP

Premium **metallic** finish with an antique look;
High strength and stiffness, excellent toughness and interlayer adhesion;
Low warpage and high precision;
Water and oil resistance, upgraded weather resistance for reliable long-term outdoor use; AMS/CFS compatibility for easy and ultra-fast printing
(up to 250 mm/s)

Applications

Designer toys, cultural and creative products, artistic ornaments, electronic DIY product enclosures, FPV drone frames, remote control car shells, planter pots, etc.

PET DET & FC Blue

USP

Most models are glue-free, detectable, and **food-contact** safe. Custom-designed for food, beverage, and pharmaceutical manufacturers, offering water and oil resistance, hygienic safety, high toughness, easy printing, support for high speeds (up to 300 mm/s), and compatibility with all major printers, AMS/CFS

Applications

Ingredient jars, flour scoops, pastry molds, and other kitchen tools; pen shells, bag clips, grab forceps, ingredient boxes, and other food plant stationery and tools; as well as food, beverage, and pharmaceutical processing equipment components — all of these items will come into contact with this product

PP GF Grey

USP

Ultra-**low density**, extremely lightweight; hydrophobic, oil and grease-resistant, and extremely resistant to chemical corrosion; high toughness and layer adhesion strength, fatigue and flex-resistant; wide service temperature range (approximately -20 °C - 100 °C); glass fiber reinforced, excellent insulation property; less warping, easier to print, compatible with mainstream printers and multi-color feeding systems such as AMS and CFS, available in various colors

Applications

Vehicle brackets and accessories for toy racing cars, drones, and robots that require lightweight, chemical reagents handling tools, electroplating fixtures, and cleaning benches that require resistance to acids, alkalis, solvents, and greases, and buckles, hinges, and movable joints that require fatigue and flex resistance, electronic component enclosures that required insulation

PP CF Black

USP

Ultra-**low density**, extremely lightweight; hydrophobic, oil and grease-resistant, and extremely resistant to chemical corrosion; high toughness and layer adhesion strength, fatigue and flex-resistant; wide service temperature range (approximately -20 °C - 100 °C); carbon fiber reinforced, smooth and delicate surface; less warping, easier to print, compatible with mainstream printers and multi-color feeding systems such as AMS and CFS

Applications

Vehicle brackets and accessories for toy racing cars, drones, and robots that require lightweight, chemical reagents handling tools, electroplating fixtures, and cleaning benches that require resistance to acids, alkalis, solvents, and greases, and buckles, hinges, and movable joints that require fatigue and flex resistance

PET GF Blue

USP

High strength, high rigidity, enhanced toughness and interlayer bonding strength, capable of bearing weight and compression; low warping, creep-resistant, higher dimensional accuracy; water and oil resistant, improved weather resistance, suitable for indoor water environments and long-term outdoor use; easy to print, compatible with AMS and CFS, supports high-speed printing (up to over 250 mm/s), available in various colors

Applications

Functional prototypes, tooling fixtures, electric tool components, electronic and electrical enclosures, high-pressure water gun housings, hose connectors, toolbox handles, etc.

PET CF Black

USP

High strength, high rigidity, enhanced toughness and interlayer bonding strength, capable of bearing weight and compression; low warping, creep-resistant, higher dimensional accuracy; water and oil resistant, improved weather resistance, suitable for indoor water environments and long-term outdoor use; easy to print, compatible with AMS and CFS, supports high-speed printing
(up to over 300 mm/s)

Applications

Functional prototypes, tooling fixtures, electric tool components, electronic and electrical enclosures, high-pressure water gun housings, hose connectors, toolbox handles, etc.

ASA GF Matte Blue

USP

Top **UV resistance**, and weather resistance, outstanding water and heat resistance, low density, high rigidity, ultra-low warpage (no need for positive chamber heating), excellent dimensional stability and surface quality, compatible with AMS/CFS, ultra-fast printing (up to 300 mm/s), multiple colors printing

Applications

Garden plant pots, camping tool boxes, patio lamps, birdhouses, swimming pool pipe fittings, weather monitoring equipment covers, ship parts, aircraft models, ship models, and other items used in outdoor and water environments or requiring lightweight and high rigidity

ASA CF Black

USP

Excellent **weather resistance**, good heat resistance, low density, low water absorption, excellent strength and impact resistance, ultra-low warpage (no need for chamber heating), outstanding dimensional stability and surface quality, compatible with AMS/CFS, and capable of ultra-high-speed printing (up to 300 mm/s)

Applications

Birdhouses, ship parts, seedling trays, aircraft models, ship models, and other outdoor functional parts used in water environments, as well as lightweight components for ship and aircraft models

PC FR HS Navy Blue

USP

Fluoride-free and **halogen-free**, ultra-thin flame retardant (0.8 mm V-0); domestic trend aesthetics, semi-transparent high toughness; excellent interlayer adhesion, ultra-low warping; ultra-fast printing (up to 350 mm/s)

Applications

Intelligent lighting, mobile power banks, network patch panels, and computer hard drive cages and enclosures, among other consumer electronics products, require top-level flame retardancy and are free of fluorine and halogens, as well as high-strength, heat-resistant components such as outdoor stove tripods

PC GF Pomegranate Red

USP

Ultra low warpage and
excellent surface quality;
Ultra-**Low water absorption**
and stable **high strength**;
Excellent Weather Resistance;
Hyper-fast printing with AMS and CFS

Applications

Surfboard brackets and pump housings;
Outdoor insulating structural parts like power enclosures

PC CF Black

USP

Ultra-low warpage
(No heated champer needed);
Ultra-Low water absorption
and excellent weather resistance;
Ultra-high layer strength and
excellent surface quality;
Hyper-fast printing
(up to 350 mm/s)

Applications

Water environment parts like surfboard fins;
Outdoor structural parts like micro wind turbines;
Precise parts like gears

CoPA HS Red

USP

Smooth surface, low friction coefficient, resistant to wear, resistant to oil and general chemical reagents, **high strength**, rigidity and toughness in dry state, ultra-high toughness in wet state, ultra-low warpage, suitable for AMS or CFS printing, 4 basic colors of black, white, gray and red

Applications

Gears, slide rails, bushings and other parts that require wear resistance, mechanical arms, drone brackets, toy racing cars, motorcycle fenders and other parts that require a balance of strength, stiffness and toughness

PA6 GF Red

USP

Glass fiber reinforced;
Ultra low warping,high heat resistance;
Superior interlayer bonding;
High toughness and impact resistance, excellent insulation; Black, white,
gray, and red;Compatible with
AMS and CFS multi-color printing

Applications

Applications need high impact and heat resistance, high strength and insulation, such as in RC cars, drone components, power tools, and hair dryer nozzles

PA6 CF Black

USP

Carbon fiber reinforced;
Ultra **low warping**,high heat resistance; Superior interlayer bonding ;
High toughness and impact resistance;
Compatible with AMS and CFS

Applications

Applications need light weight, high impact and heat resistance,high strength and high toughness, such as oven trays and racks, automobiles, drones, RC cars, and power tools

PA HT GF Red

USP

Glass fiber reinforced
Ultra **low warping**, strong, stiff and tough;
Highly accurate and wear-resistant;
Superb layer strength;
Outstanding heat resistance;
Compatible with AMS and CFS

Applications

Applications with lightweight, insulating
and high-performance needs such as automobiles, electrical tools and etc.

PA HT CF Black

USP

Carbon fiber reinforced
Ultra **low warping**, strong, stiff and tough;
Highly accurate and wear-resistant;
Superb layer strength;
Outstanding heat resistance;
Compatible with AMS and CFS

Applications

Applications with lightweight and high-performance needs such as automobiles, drones, RC racing cars, and
electrical tools

PPA Black

USP

High strength and toughness;
Excellent resistance to
temperature and moisture;
Ultra-low warpage and anti-creep;
High surface precision and wear resistance;
Four basic color series: black,
white, gray, and red;
Compatible with AMS and CFS

Applications

Bike parts, insulating or wear resistant parts for electronic devices and smart home appliances

PPA GF Black

USP

Balance of rigidity and
toughness with **Insulation**;
Excellent resistance to
temperature and moisture;
Ultra-low warpage and anti-creep;
Four basic color series: black,
white, gray, and red

Applications

Bike parts, insulating parts for electronic devices and smart home appliances,metal parts replacement for automotive and machinery Industries

PPA CF Black

USP

Lightweight, ultra-strong,
super-stiff, metal replacement;
Excellent resistance to
temperature and moisture;
Ultra-low warpage and anti-creep

Applications

ESD parts for the semiconductor and oil and gas industries,metal parts replacement for automotive, drone and machinery Industries

PPA GF FR Black

USP

Premium **flame retardancy**,
insulation & heat resistance;
High rigidity, low warpage,
ultra-low moisture absorption;
No chamber heating required,
widely compatible with a wide
range of printer models;
Classic four colors: black,
white, orange, and blue

Applications

Flame-retardant junction boxes, socket components, insulating fasteners, and other electronic and electrical prototypes and parts; heat-resistant tools and fixtures such as assembly jigs and inspection tooling for production lines; high-demand components for office equipment and smart hardware

PPS GF Grey

USP

Glass fiber reinforced
with top flame retardancy,
heat resistance & insulation;
High rigidity & strength,
water and chemical resistance;
Optimized interlayer adhesion,
widely compatible with
high-temperature printers;
AMS/CFS compatible for easy printing

Applications

High-strength industrial components such as motor mounts and electric skateboard drive gears; extremely high-temperature resistant insulating parts such as automotive turbo intake pipes, new energy vehicle battery mounts, and high-temperature electrical equipment enclosures; water and chemical resistant components such as scalpel handles and chemical pumps

PPS CF Black

USP

Carbon fiber reinforced
with top flame retardancy;
Extreme resistance to water,
high temperature, and solvents;
Optimized interlayer adhesion;
Wide compatibility with
high-temperature printers;
AMS/CFS compatibility for easy printing

Applications

High-strength industrial components, such as motor cooling fans, automotive heater rotors, electric skateboard gears, and CNC motor mounts; flame-retardant parts, such as turbo intake pipes; and water and high-temperature resistant components, such as motorcycle fuel tanks

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Difficulty is a reference rating; * = estimated by family. Nozzle chips: black = supported, blue = recommended. Cell tint = relative level; ▲/▼ = best (not shown for hardness/MFI). Layer-adhesion bar derived from values; Shore A/D scales are not directly comparable.

Bar length = relative level (by value); colour = row theme; thin bar = relative level   Download TDS ↗